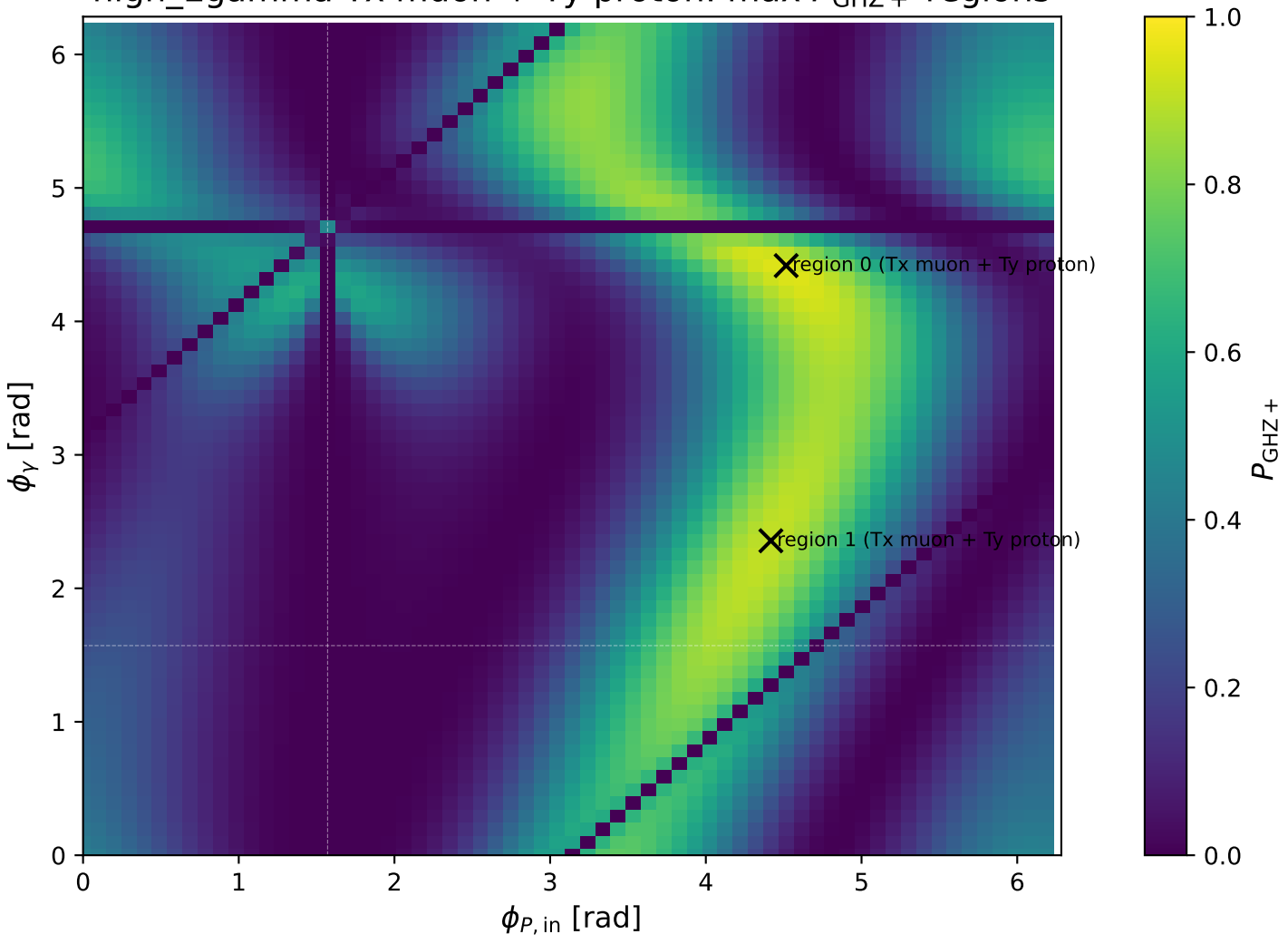
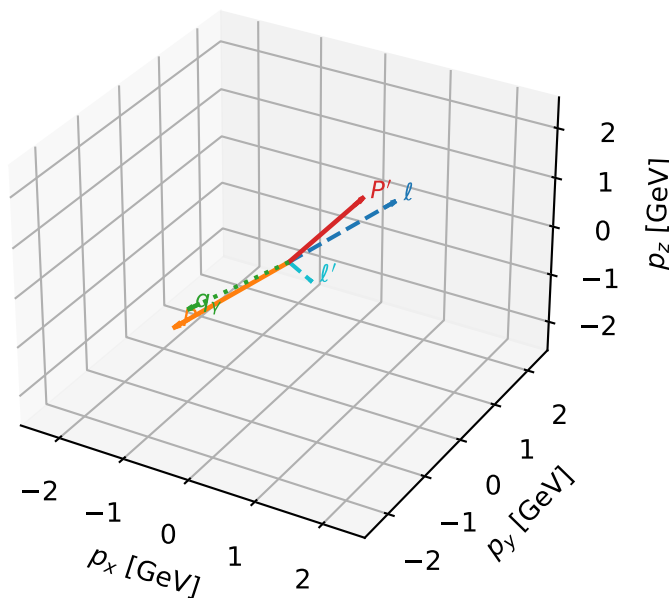


high_Egamma Tx muon + Ty proton: max $P_{\text{GHZ}+}$ regions



Tx muon + Ty proton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

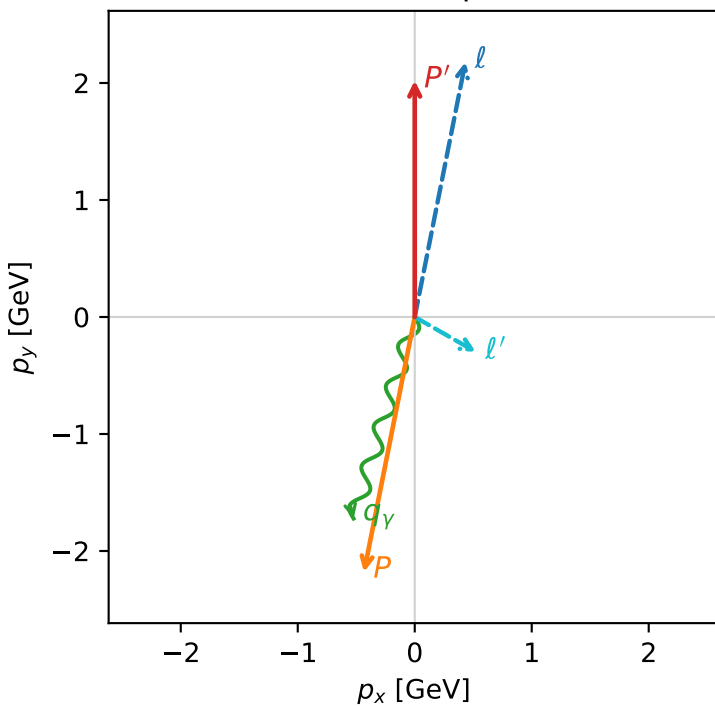


ghz_purity_high_egamma_txtty_region_0 $P_{\{\text{rm GHZ+}\}}=0.948063$
region: high_Egamma TxTy_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.02062$
 $\theta_{\text{in}}=1.57079$, $\phi_\ell=1.37445$, $\phi_P=4.51604$
 $E_\gamma=1.8$, $\phi_\gamma=4.41786$
 $Q^2=3.5433$, $x_B=0.19128$, $t=-17.8024$, $W^2=15.8607$, $y=0.898147$
 $\theta(\ell', \gamma)=77.1671$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= 0.4337$ $p_y= 2.1804$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.4337$ $p_y= -2.1804$ $p_z= 0.0000$
 ℓ' $E= 0.6108$ $p_x= 0.5225$ $p_y= -0.2981$ $p_z= 0.0000$
 P' $E= 2.2277$ $p_x= 0.0000$ $p_y= 2.0206$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -0.5225$ $p_y= -1.7225$ $p_z= 0.0000$

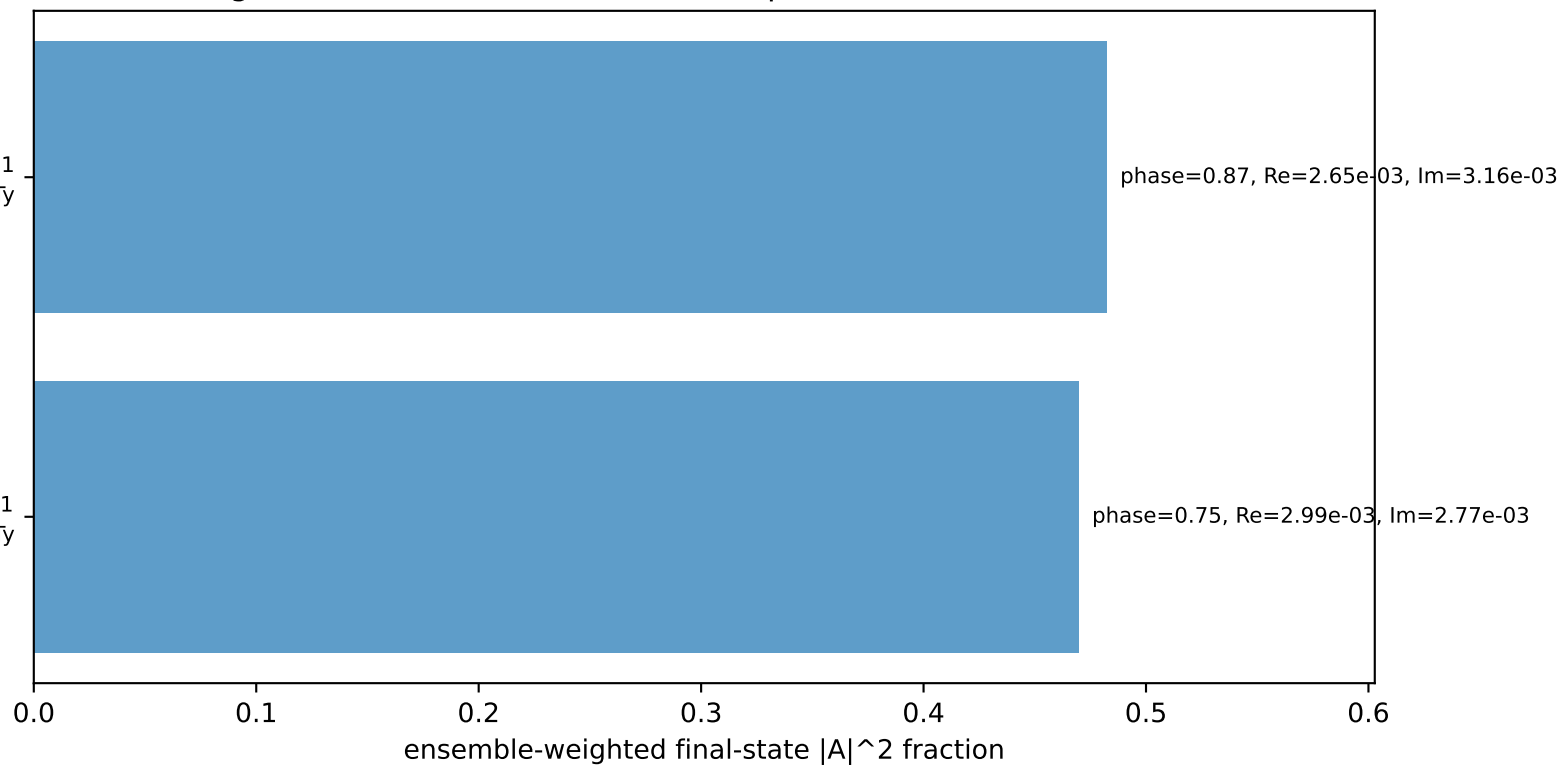
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
electron Tx, proton Ty

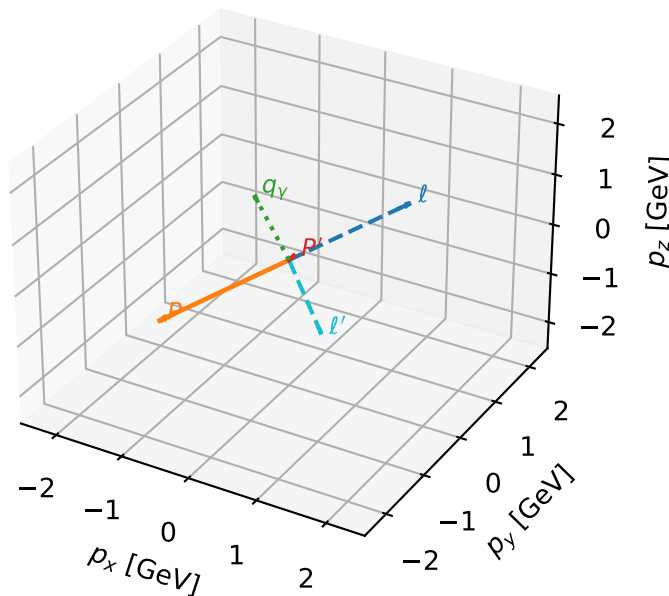
$h_e=-1, h_p=-1, h_\gamma=-1$
electron Tx, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=95.2%)



Tx muon + Ty proton characteristic max P_{GHZ} + configuration

3D momenta

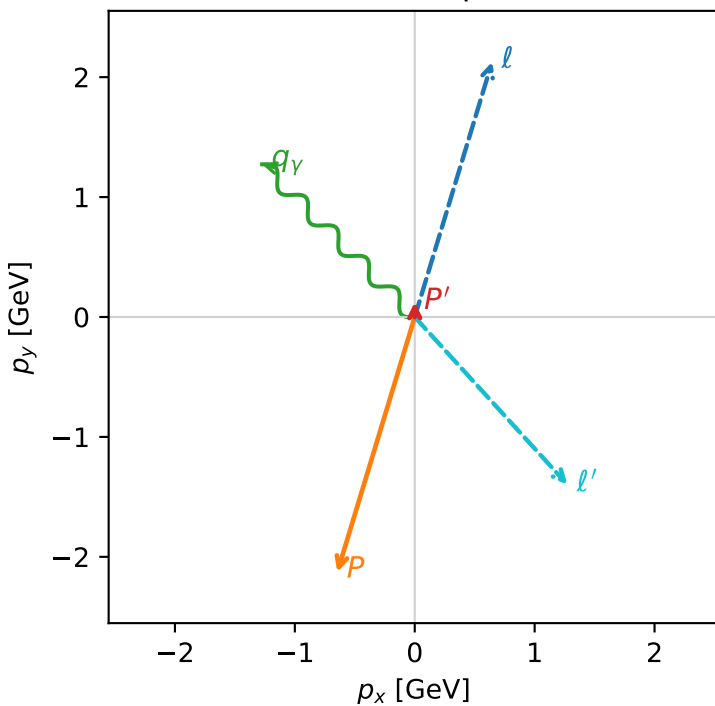


ghz_purity_high_egamma_txyt_region_1 $P_{\text{GHZ}}=0.907354$
 region: high_Egamma TxTy_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|T_{x,\mu}\rangle \otimes |\bar{T}_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.123642$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.27627$, $\phi_p=4.41786$
 $E_\gamma=1.8$, $\phi_\gamma=2.35619$
 $Q^2=12.7$, $x_B=0.804242$, $t=-3.33213$, $W^2=3.97111$, $y=0.765642$
 $\theta(l', \gamma)=177.348$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 0.6453$ $p_y= 2.1274$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.6453$ $p_y= -2.1274$ $p_z= 0.0000$
 l' $E= 1.8924$ $p_x= 1.2728$ $p_y= -1.3964$ $p_z= 0.0000$
 P' $E= 0.9461$ $p_x= 0.0000$ $p_y= 0.1236$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -1.2728$ $p_y= 1.2728$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=96.2%)

